

(TR-104) PYTHON WITH AI/ML

Training Day 62 Report

Date: 7 April 2026

Topic Covered: Multi-LLM Integration, Guardrails & Tool Call Validation

The sixty-second day of training focused on building a unified multi-model AI system supporting different LLM providers, implementing safety guardrails, and enforcing structured tool-call validation. The session emphasized making the system robust, secure, and production-ready for multi-provider environments.

- Built a universal LLM router to support multiple providers including Ollama, Hugging Face, and Gemini.
- Implemented provider-based routing logic using a centralized `call_llm` function.
- Integrated Ollama local API for fast and efficient local model inference.
- Added Hugging Face Inference API support with secure token-based authentication.
- Integrated Gemini API using OpenAI-compatible request structure.
- Standardized structured response parsing across different LLM providers for consistency.
- Implemented tool call argument normalization to handle provider-specific response formats.
- Built prompt injection detection system using LLM-based validation techniques.
- Developed content safety checker to filter unsafe or malicious outputs.
- Designed robust system prompts for moderation and secure response generation.
- Added fallback handling mechanisms for API failures and invalid responses.
- Implemented JSON schema validation using `jsonschema` for tool call enforcement.
- Ensured strict validation of tool arguments against predefined schemas.
- Handled malformed or invalid tool inputs gracefully to prevent system crashes.

- Added parsing logic to convert string-based tool arguments into structured JSON format.
- Suggested corrections and improvements in project documentation including AI tools naming consistency (GitHub Copilot, LlamaIndex, Claude, etc.).
- Strengthened understanding of multi-LLM orchestration, safety guardrails, and structured tool-calling systems in production-grade AI applications.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/workflows>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/model-context-protocol-mcp-connecting-ai-to-the-world-be9b49f1394f>